OPERATING SYSTEMS

UNIT I – OPERATING SYSTEM BASICS

SECTION A – 2-MARK QUESTIONS (3-MARK ANSWER FORMAT)

Answers, examples and simple diagrams

Q1. Define batch processing system.

Definition:

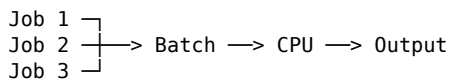
A Batch Processing System is an operating system method that collects similar jobs and executes them together in a group called a batch.

Explanation:

- Jobs are collected first.
- The OS executes jobs one by one without user interaction.
- It saves time by processing similar jobs together.

Example: Processing electricity bills of many customers together.

Diagram:



Q2. What are Layered Operating Systems?

Definition:

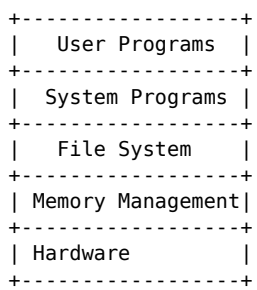
A Layered Operating System is an OS divided into different layers. Each layer performs a specific task.

Explanation:

- Each layer has its own function.
- The upper layer uses the services of the lower layer.
- It makes the OS easy to design and maintain.

Example: Windows operating system.

Diagram:



Q3. What is an Operating System?

Definition:

An Operating System (OS) is system software that acts as a bridge between the user and computer hardware.

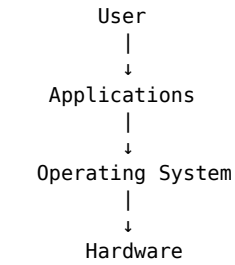
Explanation:

- It manages computer resources.

- It controls hardware and software.
- It helps users run different applications.

Example: Windows, Linux, Android.

Diagram:



Q4. What are System Calls in Operating Systems?

Definition:

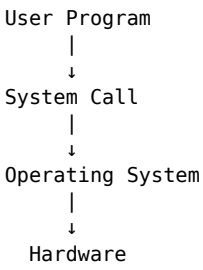
System Calls are the methods used by a program to request services from the Operating System.

Explanation:

- They help programs communicate with the OS.
- They request services like file creation and reading.
- They provide a connection between user programs and the OS.

Example: read(), write(), open().

Diagram:



Q5. List any two functions of an Operating System.

Definition:

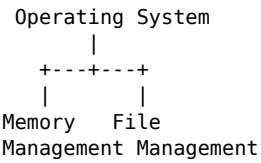
An Operating System performs different functions to manage a computer system.

Explanation:

- Memory Management: It manages the main memory and allocates memory to programs.
- File Management: It creates, stores, deletes and organizes files.

Example: Windows manages RAM and stores files in folders.

Diagram:



Q6. What is the difference between Open-Source and Proprietary Operating Systems?

Definition:

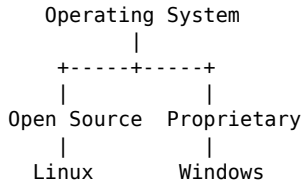
Open-source OS allows users to view and modify its source code, whereas Proprietary OS does not allow users to freely modify its source code.

Differences:

Open-Source OS: Source code is available; users can modify it; Example: Linux. Proprietary OS: Source code is not freely available; users cannot freely modify it; Example: Windows.

Example: Open-source: Linux. Proprietary: Windows.

Diagram:



Q7. Define Kernel.

Definition:

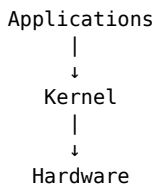
The Kernel is the main part of an Operating System. It acts as a bridge between software and hardware.

Explanation:

- It manages CPU and memory.
- It controls input and output devices.
- It manages running processes.

Example: When we open an application, the kernel helps the CPU execute it.

Diagram:



Q8. What is Multiprogramming?

Definition:

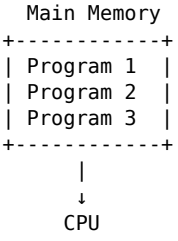
Multiprogramming is a technique in which multiple programs are kept in memory at the same time to keep the CPU busy.

Explanation:

- Many programs are loaded into memory.
- When one program waits for I/O, the CPU executes another program.
- It increases CPU utilization.

Example: While one program waits for keyboard input, the CPU can execute another program.

Diagram:



Q9. How does Dual-Mode Operation work?

Definition:

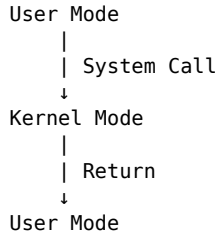
Dual-Mode Operation is a technique that provides two modes of operation to protect the computer system.

Explanation:

- User Mode: Programs run with limited permissions.
- Kernel Mode: The OS runs with full permissions.
- When a program needs an OS service, it requests a system call to switch to kernel mode.

Example: A normal application runs in user mode, but file access may require OS services.

Diagram:



Q10. What are the objectives of Operating Systems?

Definition:

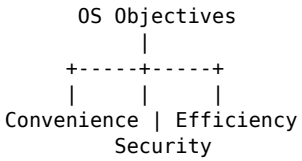
The main objectives of an Operating System are to make the computer easy to use and manage its resources efficiently.

Explanation:

- Convenience: Makes the computer easy to use.
- Efficiency: Uses CPU and memory properly.
- Security: Protects data and programs.

Example: Android makes mobile phones easy and safe to use.

Diagram:



Q11. What is the use of Fork and Exec System Calls?

Definition:

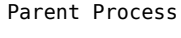
Fork and Exec are system calls used to create and execute processes in an Operating System.

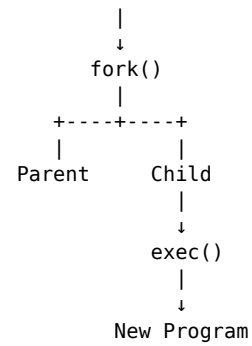
Explanation:

- Fork(): Creates a new process from an existing process.
- Exec(): Replaces the current process with a new program.
- Both are commonly used in UNIX/Linux systems.

Example: A parent process creates a child process using fork(), and the child runs another program using exec().

Diagram:





Q12. Define Open-Source Operating System.

Definition:

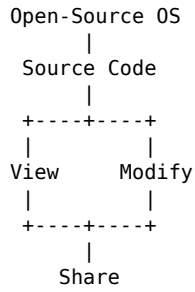
An Open-Source Operating System is an OS whose source code is freely available for users to view, modify and share according to its license.

Explanation:

- Users can view its source code.
- Developers can modify and improve it.
- It allows sharing according to its license.

Example: Linux, Ubuntu.

Diagram:



Q13. List the services provided by an Operating System.

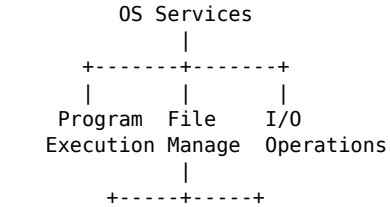
Definition:

Operating System Services are the facilities provided by the OS to users and programs.

Explanation:

- Program Execution: Runs programs.
- I/O Operations: Controls input and output devices.
- File Management: Creates and manages files.
- Communication: Allows processes to communicate.
- Error Detection: Detects system errors.
- Security: Protects data and programs.

Diagram:



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    |          |
Communication Security
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Q14. What is the purpose of System Programs?

Definition:

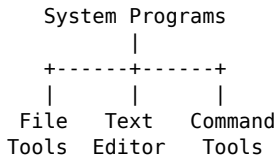
System Programs are utility programs that provide a convenient environment for developing and executing programs.

Explanation:

- They help users perform common computer tasks.
- They manage files and directories.
- They help in program development and execution.

Example: File Explorer, Command Prompt, Text Editors.

Diagram:



Unit I – Quick Revision

Question	Important Keywords
Q1	Jobs, Batch, Together
Q2	Layers, Specific Tasks
Q3	Bridge, User, Hardware
Q4	Request, OS Services
Q5	Memory, File Management
Q6	Source Code, Linux, Windows
Q7	Main Part, Hardware
Q8	Multiple Programs, CPU
Q9	User Mode, Kernel Mode
Q10	Convenience, Efficiency, Security
Q11	Fork = Create, Exec = Replace
Q12	View, Modify, Share
Q13	Execution, I/O, Files, Security
Q14	Utility Programs, Environment